

Application of ICT Database systems

Lecture 12 & 13



What Is a Database?

- A **Database** is an organized collection of **data** that can be **easily accessed, managed, and updated**.
 - **Data**: Raw facts (e.g., “Ali”, “101”, “A Grade”)
 - **Information**: Processed data that has meaning (e.g., “Ali scored A Grade in English.”)
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Database

A database is a structured collection of data that allows people to extract information in a way that meets their needs. The data can include text, numbers, pictures; anything that can be stored in a computer.

Example

Student_ID	Name	Class	Marks	Contact
101	Ali	8	A	0301-XXXXXXX
102	Sara	9	B	0302-XXXXXXX

How Databases Improve Information Management

Traditional Method

Paper files or spreadsheets

Data duplication and loss

Difficult to search

Manual updates

No access control

Using a Database

Digital records

Centralized, secure data

Instant searching and reporting

Automatic and accurate updates

User-based permissions and backups



Why are databases useful?

- Databases prevent problems occurring because:
 - » if any changes or additions are made it only has to be done once – data is Consistent.
 - » the same data is used by everyone.
 - » Data is only stored once in relational databases which means no data duplication.
-



What are databases used for?

- **To store information about people, for instance:**
 - » patients in a hospital
 - » pupils at a school.
 - **To store information about things, for instance:**
 - » cars to be sold
 - » books in a library.
 - **To store information about events, for instance:**
 - » hotel bookings
 - » results of races.
-



Fields and records – the building blocks for any database

Inside a database, data is stored in **tables**, which consists of many **records**.

Each record consists of several **fields**.

The number of records in a table will vary as new records can be added and deleted from a table as required.

The number of fields in a table is fixed so each record contains the same number of fields.

An easy way to remember this is: each **record is a row** in the table and each **field is a column** in the table.

Real-Life Examples of Database Use

Sector	Example of Database
Education	Student record systems, attendance databases
Healthcare	Patient management system, doctor schedules
Banking	Customer accounts, transaction records
E-commerce	Product catalog, customer orders
Government	National ID, tax and census databases



Discussion

How does your school or a hospital keep track of all student or patient information?

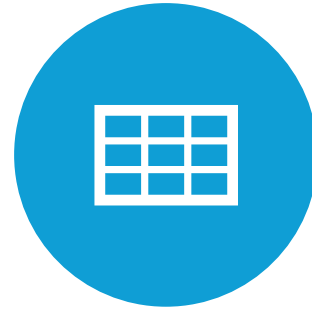
Expected Answer



REGISTERS



FILES AND FOLDERS



EXCEL SHEETS



SOFTWARE OR
DATABASES

Task: Identify Data Items for a Hospital Database

Design a **Hospital Management System**.
Identify **5 key data items** you would store for each patient.

Data Items



Patient_ID



Patient_Name



Age



Disease_Diagnosis



Contact_Number



What kind of information could be derived from this data?

- For instance: number of patients with a certain disease, or average age per ward
-



Database Management Systems

- A **Database Management System (DBMS)** is software that helps us **store, organize, and manage data efficiently**. But a DBMS is more than just software — it includes **five key components** that work together like parts of a machine.
 - These components are:
 - **Hardware**
 - **Software**
 - **Data**
 - **Users**
 - **Procedures**
-



Scenario: Online Shopping System

Let's imagine an **online shopping store** called **ShopSmart.pk**.

This company sells clothes and electronics online. Every time a customer places an order, many things happen behind the scenes in the database.

1. Hardware(servers, laptop, computers or mobile devices)
 2. Software (MYSQL,PHP, Python)
 - DBMS software (MySQL, Oracle, MS Access)
 - Operating system
 - Application programs that access the database
-

3. Data

Table	Example Data
Customers	Customer_ID, Name, Email, Address
Products	Product_ID, Name, Price, Stock
Orders	Order_ID, Date, Customer_ID, Payment_Status

4. Users

User Type	Role
Database Administrator (DBA)	Manages the database, creates backups, controls access
Application Programmer	Writes code that connects the website to the database
End Users (Customers)	Browse products and place orders using the website
Data Analyst	Analyzes customer buying patterns and sales reports

5.Procedures



The instructions, rules, and processes that guide how data is collected, stored, and managed.



At ShopSmart.pk, procedures include:



How new products are added to the database



How customer data is updated



Backup and recovery policies



Security rules (like password protection and encryption)



Database Software

- Used to store and manage data.
 - Features:
 - Tables, queries, reports, forms
 - Searching & filtering
 - Multi-user access
 - Examples:
 - MS Access, MySQL, Oracle
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Database

- Data in a database is organized into fields (columns), records (rows), and tables
 - **Field (column)**
 - A single type of data to be stored in a database
 - **Record (row)**
 - A collection of related fields
 - **Table**
 - A collection of related records
 - **Database file**
 - Collection of related tables
-



Entity

An entity in a database is a thing, place, person or object that is independent of another.





Database Table

- Table structure includes:
 - Field name (unique identifying name)
 - Data type (text, number, date, object)
 - Field size (maximum number of characters)
 - Default value (initial content of the field)
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Table (Reference Book)

Table

The diagram illustrates the structure of a database table. It features a table with 6 rows and 4 columns. The first column lists records from 'Record 1' to 'Record 6'. The subsequent columns are labeled 'Field 1', 'Field 2', 'Field 3', and 'Field 4'. A callout box labeled 'Row' points to the second row (Record 2). Another callout box labeled 'Column' points to the fourth column (Field 4). The cells for 'Record 2' and 'Field 4' are highlighted in light blue.

Record 1	Field 1	Field 2	Field 3	Field 4
Record 2	Field 1	Field 2	Field 3	Field 4
Record 3	Field 1	Field 2	Field 3	Field 4
Record 4	Field 1	Field 2	Field 3	Field 4
Record 5	Field 1	Field 2	Field 3	Field 4
Record 6	Field 1	Field 2	Field 3	Field 4

▲ **Figure 9.1** Structure of a database table

A table contains data about one type of item or person or event, and will be given a meaningful name, for example:

- » a table of patients called **PATIENT**
- » a table of books called **BOOK**
- » a table of doctor's appointments called **APPOINTMENT**.

Each record within a table contains data about a single item, person or event, for example:

- » Winnie Sing (a hospital patient)
- » IGCSE Computer Science (a book)
- » 15:45 on January 2020 (an appointment).

As every record contains the same number of fields, each field in a record contains a specific piece of information about the single item, person or event stored in that record. Each field will have a meaningful name to identify the data stored in it, for example:

- » For a hospital patient the fields could include:
 - The patient's first name field called **FirstName**
 - The patient's family name field called **FamilyName**
 - The patient's date of admission field called **DateOfAdmission**

- The name of the patient's consultant field called **Consultant**
- The patient's ward number field called **WardNumber**
- The patient's bed number field called **BedNumber**, etc.

PATIENT Table

Record 1	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 2	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 3	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 4	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 5	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 6	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber

Activity

State what fields would you expect to find in each record for the doctor's appointments and give each field a suitable name.

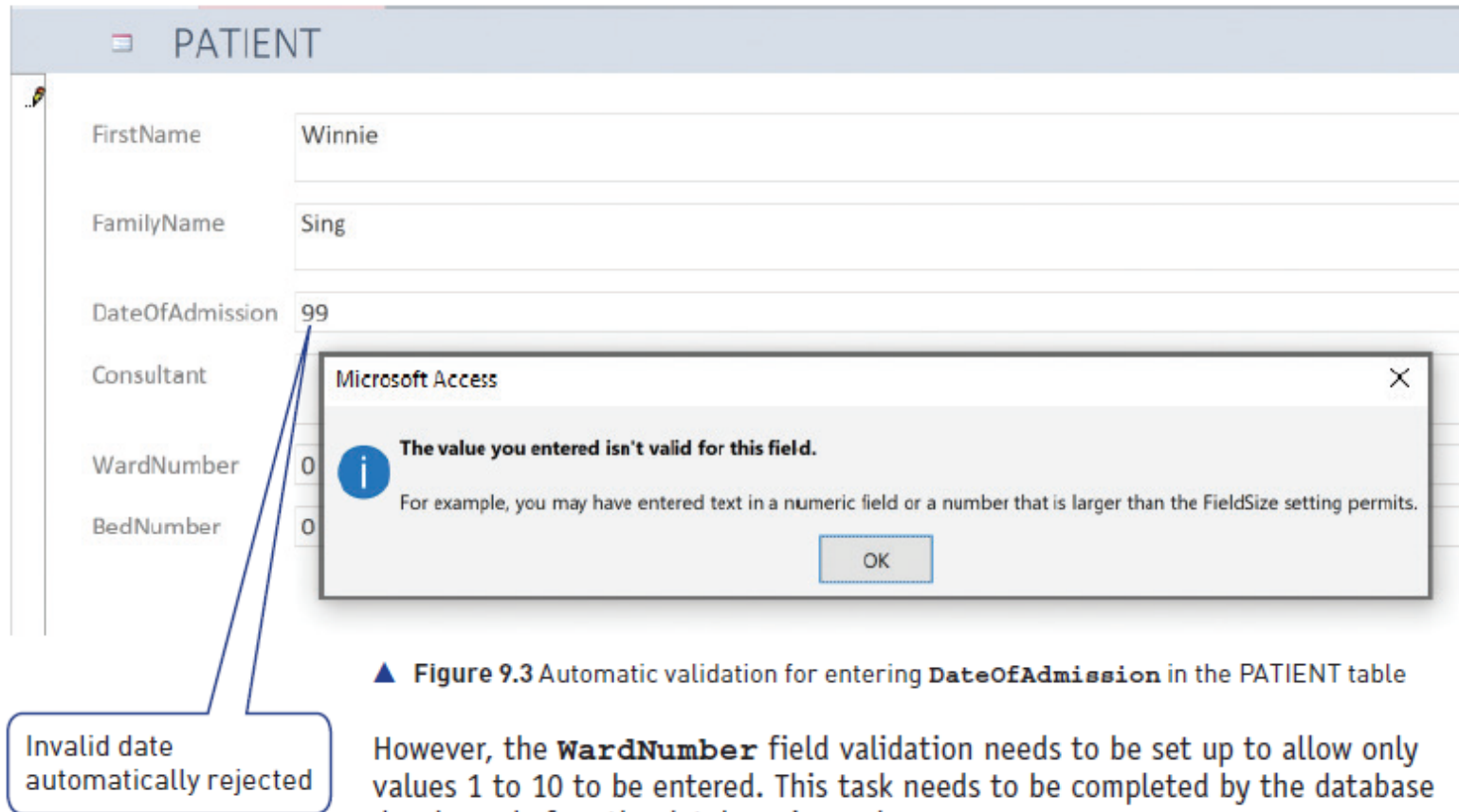
Note: Field names should be a single word, which should not contain any spaces, for example: **BedNumber**

Validation

- Some validation checks will be automatically provided by the database management software that is used to construct and maintain the database. Other validation checks need to be set up by the database developer during the construction of the database.

The **DateOfAdmission** field will automatically be checked by the software to make sure that any data input is a valid date before it can be stored in the .PATIENT table

Database



Database

Out of range ward number rejected

General	
Field Size	Integer
Format	
Decimal Places	0
Input Mask	
Caption	
Default Value	0
Validation Rule	≥ 1 And ≤ 10
Validation Text	Wards numbers are 1 to 10
Required	No
Indexed	No
Text Align	General

PATIENT

FirstName: Winnie

FamilyName: Sing

DateOfAdmission:

Consultant:

WardNumber: 67

BedNumber: 0

Microsoft Access

! Wards numbers are 1 to 10

OK Help

Error message

▲ Figure 9.4 Validation rule for entering **WardNumber** in the PATIENT table

012 Basic data types



Tools

- Application software helps users perform specific tasks.
 - Three types: General-purpose, special-purpose, custom.
 - Productivity tools:
 - Word Processing → Documents
 - Spreadsheets → Data analysis
 - Databases → Data management
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Primary keys

- As each record within a table contains data about a single item, person, or event, it is important to be able to uniquely identify this item. In order to reliably identify an item from the data stored about it in a record there needs to be a field that uniquely identifies the item. This field is called the **primary key**.
- A field that is a primary key must contain data values that are never repeated in the table.

Basic data types

- There are six basic data types that you need to be able to use in a database:
 - » text/alphanumeric
 - » character
 - » Boolean
 - » integer
 - » real
 - » date/time.

What is a data type?

- Each field will require a **data type** to be selected. A data type classifies how the data is stored, displayed and the operations that can be performed on the stored value.
- For example, a field with an integer data type is stored and displayed as a whole number and the value stored can be used in calculations.

Activity

- 1 Write down **four** database data types. Describe each data type and give an example of a field where it would be a suitable choice.
- 2 Choose suitable words/phrases from the following list to correctly complete the paragraph that follows:

Word list: database fields records
 record table primary key
 data type validation text
 char Boolean field

You will need to use some words more than once.

A single contains one

Each consists of many Every

has the same number of Every is given a

..... Examples of data types are,

and Some will have rules.

Data Type	Description	Example Field
Text (String)	Used to store alphanumeric characters (letters, numbers, symbols).	CustomerName = "John Smith"
Integer (Number)	Used to store whole numbers without decimals.	Age = 25
Boolean	Stores TRUE/FALSE or YES/NO values.	IsActive = TRUE
Date/Time	Used to store calendar dates and times.	OrderDate = 23/10/2025



Q2

- **Word list:** database, fields, records, record, table, primary key, data type, validation, text, char, Boolean, field
- A single **database table** contains one **primary key**.
Each **table** consists of many **records**.
Every **record** has the same number of **fields**.
Every **field** is given a **data type**.
Examples of data types are **text**, **char**, and **Boolean**.
Some **fields** will have **validation** rules.

For further reading | Additional resources

- Chapter 9 of the book Titled “Computer Science Second Edition by David Watson and Helen Williams”

